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An Engineering Guide to Quadrotor Pursuit of Fast Fixed-Wing Aircraft

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Abstract

A quadrotor must keep a fast fixed-wing aircraft in the image of one camera in each frame. The aircraft is 3.5 m long, flies at speeds up to 55 m/s and turns on a radius of 350 m. A computer has checked all the bounds that we use. One bound is sufficient: if it is true, our pursuit method keeps the target in view. The other bound is necessary: if it is false, no method can keep the target in view. We changed each design parameter and recorded the effect on the two bounds.

The results give three rules. Thrust and air set the top speed, and the thrust-to-weight ratio is the most effective parameter. Yaw authority sets the agility in a dive at mid speed, and more thrust does not help there. The camera position solves the camera problem: fly below the target and point a hinged camera up and back. More complex pursuit methods did not give better results than a simple copy of the target path.

For a 1.5 kg quadrotor, the proved limit is 57.7 m/s, and our proofs certify our method to 52 m/s. A recommended set of changes increases these values to 73.9 m/s and 68 m/s.

1 Introduction

A quadrotor that follows a fast fixed-wing aircraft must keep the aircraft in view at all times. This looks like a problem of control. Our results show that it is almost always a problem of hardware.

We made a full rigid-body model of the quadrotor and of the target. In the Lean 4 proof assistant [1, 2], we proved two inequalities about this model. The first inequality is sufficient for one pursuit method. The second inequality is necessary for all pursuit methods.

Then we changed each physical parameter and recorded how the two inequalities moved. This guide gives the results in a form that an engineer can use. It tells you which parameters to change, by how much, and which parameters to leave alone.

Our reference quadrotor has a mass of 1.5 kg and a thrust-to-weight ratio of 6. It has four motors in a plus layout, on arms of 18 cm. Its yaw reaction coefficient is 0.02 m, and its inertia is $(0.012, 0.012, 0.02)$ kg m^2.

Its parasitic drag coefficient c is $0.008\,\mathrm{m}^{-1}$. The rotor disc also has a normal-force coefficient k_n of $0.01\,\mathrm{m}^{-1}$. This value agrees with approximately $0.02\,\mathrm{m}^2$ of normal area. The tilt limit is 85°, and the body-rate limit is 20 rad/s.

A camera with a 120° field of view is on the top of the quadrotor. The camera points to the rear, on a hinge with a range of 30° to 70°. The control system operates at 240 Hz.

The reference target flies at speeds from 20 m/s to a maximum speed $v_{\max}$. Its minimum turn radius is 350 m. Its acceleration along the path is not more than $2\,\mathrm{m/s}^2$. Its vertical acceleration is not more than $1.5\,\mathrm{m/s}^2$. Its snap is not more than $15\,\mathrm{m/s}^4$, and its jerk is between 1 and $5\,\mathrm{m/s}^3$. It can climb or dive at angles of up to 5° or 10°.

Table 1 gives the answer on one page. The quadrotor gets the most speed from more thrust and from a smaller rotor-disc area. It gets the most agility in steep dives from more yaw authority and less yaw inertia. A hinge or a wide lens removes the camera as a limit. A longer arm, a cleaner airframe, wings and a more complex planner do not help. The subsequent sections explain each of these results.

Table 1: Summary for the reference quadrotor. The limit is the proved necessary bound. The certified value is the highest $v_{\max}$ for which the proved sufficient condition is true.

Question	Answer
Highest target speed that any method can follow (proved limit)	57.7 m/s
Certified speed, dives to 5°, jerk to 5	52.1 m/s
Certified speed, dives to 10°, jerk to 3	52.1 m/s
Certified speed, dives to 10°, jerk 5	33.4 m/s (yaw limit)
with yaw coefficient 0.03 or 30% less inertia	52.1 m/s
Recommended set of changes (Section 8)	limit 73.9; certified 68.1 and 58.4
Limit at top speed	the top motor is at maximum thrust
Limit in dives	yaw torque at low total thrust
Is the camera a limit?	no, with a 120° lens or a hinge

2 How to read the numbers

This guide uses two types of bound. The necessary bound is a speed limit. If the target flies faster than this limit, a straight and level target escapes from all quadrotor paths. This is true also for a path that the quadrotor plans with full knowledge of the target's future. The Lean theorems are `master_necessary_power` for the rotor-disc model and `master_necessary3` for the isotropic model.

The sufficient bound is the inequality `MasterIneq3`. If it is true, our method keeps each permitted target in view in each frame. The target must start in steady flight. The Lean theorem is `master_sufficient3`. The certified speed is the highest $v_{\max}$ for which `MasterIneq3` is true. A numerical copy of the Lean definitions finds this value.

The model is a full rigid-body quadrotor in discrete time, with a step of $1/240$ s. It includes four motors with limits, the motor mixer and the rotational equation of motion. It also includes exact rotations and the limits on tilt, body rate and speed. Air causes two forces. The first force is parasitic drag, $-c\|v\|v$. The second force is a normal force on the rotor disc [3, 4]:

$$a_n = -k_n \|v\| \langle v, b_3 \rangle \, b_3. \tag{1}$$

Here v is the velocity and b_3 is the thrust axis. This force acts at the centre of mass. The target is in view when it is in the detection range and in the field of view.

The model does not include five effects. These effects are the change of motor thrust with airspeed, aerodynamic moments, wind, the ground and sensor noise. Thus each number in this guide is true for the model only. Section 9 tells you what to measure before you use these numbers.

3 Geometry: fly below the target and look back and up

3.1 Why the quadrotor must fly below the target

A fast quadrotor tilts a large amount. The pitch angle is approximately 50° at 35 m/s and 70° to 80° at 55 m/s. A camera below the body, with the quadrotor above the target, cannot keep the target in view above approximately 30 m/s. At these speeds, the tilt of the thrust axis moves the target out of the field of view.

A position above the target is also a bad position for a different reason. To look down, the quadrotor must pitch more than its drag balance. This extra pitch pushes the quadrotor forward. The quadrotor then passes the target and loses it. The theorem `QuadTraj.forced_loss_time` gives the time until this loss.

In a prototype with a range of 382 m and a 90° lens, the time was approximately 17 s. This was at 20 m/s, 300 m above the target. At 30 m/s, the time was approximately 53 s.

Below the target, the opposite occurs. When the quadrotor pitches its nose down for speed, a rear camera on its top turns up. Thus, speed helps this camera.

3.2 Point the camera to the rear

On a quadrotor with its nose down, a forward camera on the top points at the angle $\eta - p$. Here p is the pitch angle and η is the camera angle. At high speed, this angle is zero or less, so the camera cannot look up.

A rear camera points at the angle $p + \eta$ from the rear horizon. At high speed, this camera points straight up. Install the camera on the rear face of a cone-shaped head. Attach the camera with a hinge that turns about the pitch axis.

3.3 Use a hinge or a wide lens

Our method locks the yaw to the side direction of the target. With this method, the camera error angle is not more than the bank angle plus the hinge error. The bank angle is not more than 38° in all conditions. The camera error is not more than 40°. Table 2 shows when the hinge is necessary.

Table 2: Certified $v_{\max}$ in m/s. Isotropic drag, dives to 5°, jerk to 3.

Camera installation	120° lens	90° lens
Hinge, 30° to 70°	69.9	69.9
Hinge, 20° to 80°	69.9	69.9
Fixed at 50°	69.9	**50.1**
Fixed at 60°	—	**34.6**

With a lens of 120° or more, a fixed camera at approximately 50° is sufficient. With a lens of 90°, use a hinge with a range of 30° to 70° or more. The pitch angle changes by approximately 80° between 20 m/s and 55 m/s. Only a hinge or a wider field of view can compensate for this change.

3.4 Vertical distance and ground clearance

The vertical distance between the quadrotor and the target has almost no effect on the bounds. Distances from 60 m to 150 m give the same results. But this distance sets the ground clearance. The target must fly higher than this distance plus a safety margin. Use a distance of 60 m to 100 m.

4 Top speed: thrust and air

4.1 The physics

At high speed, the rotor disc is almost perpendicular to the airflow. At 55 m/s, its tilt is approximately 68°. Thus, the normal force on the disc is large. It is approximately equal to the thrust that the quadrotor uses against parasitic drag.

Let F be the total thrust per unit mass, with $F \le F_{\max}$. Let $y = \langle v, b_3 \rangle$. The power that the motors can put into the flight path is then limited:

$$F y - k_n \|v\| y^2 \;\le\; \min\Big(F_{\max}\|v\|,\; \frac{F_{\max}^2}{4 k_n \|v\|}\Big). \tag{2}$$

Parasitic drag removes the power $c\|v\|^3$. The quadrotor must also stay in camera range of a level target, so its height cannot change without limit. The energy balance then gives the proved speed limit (`no_traj_keeps_of_power`). To the first order, this limit is

$$V_{\text{limit}} \approx \Big(\frac{F_{\max}^2}{4\, k_n\, c}\Big)^{1/4}. \tag{3}$$

4.2 Effect of each parameter

Table 3 shows the limit and the certified speed for each parameter value. Equation (3) agrees closely with the data. When $F_{\max}$ doubles, the limit increases by a factor of 1.42. The equation gives $\sqrt{2}$, which is 1.41. When k_n increases by a factor of four, the limit decreases by a factor of 0.71. The equation gives 0.707.

When c increases from 0.006 to 0.010, the limit decreases by a factor of 0.88. The equation gives the same value. The certified speed is approximately 90% of the limit in all cases.

Table 3: Proved limit and certified $v_{\max}$ in m/s. Dives to 5°, jerk to 5. All other parameters have their reference values.

Parameter	Value	Limit	Certified
Thrust-to-weight ratio	4	47.1	37.6
	5	52.6	46.2
	6	57.7	52.1
	7	62.3	57.3
	8	66.7	61.9
k_n (m^{-1})	0	84.5	69.5
	0.005	68.4	61.6
	0.01	57.7	52.1
	0.02	48.7	40.8
Drag c (m^{-1})	0.006	62.0	39.3
	0.008	57.7	52.1
	0.010	54.5	50.1

4.3 How close the limit is to the physics

The limit of 57.7 m/s is a physical limit. In steady, straight and level flight at maximum thrust, the quadrotor gets to 57.3 m/s. The difference of 0.4 m/s comes from the energy gain of the time step at 240 Hz.

A point mass can fly faster than steady flight with a weave maneuver. It applies more thrust than necessary and turns the extra thrust to the left and to the right in turn. The side force does no work, but it tilts the rotor disc away from the airflow. This decreases the drag on the disc. At $k_n = 0.02$, the weave gives 47.8 m/s, and steady flight gives 47.0 m/s.

A real quadrotor cannot fly this weave. For the fast changes of bank angle, motor thrusts from −34 N to +78 N are necessary. The motor limit is 22 N. Also, this torque is necessary when the total thrust is already at maximum. In our search, each weave within the limits was slower than steady flight.

4.4 Rules for top speed

Increase the thrust-to-weight ratio first. It is the only parameter that increases both the limit and the certified speed by approximately 5 m/s per unit.

Then decrease the area of the rotor plane that faces the airflow. Use small rotor discs for the thrust. Use a thin frame in the plane of the discs.

Do not decrease the parasitic drag too much. A lower c increases the limit only as $c^{-1/4}$, and it decreases the agility in dives. Keep a thrust reserve on each motor at top speed. The top motor gets to its maximum thrust first, because it supplies thrust for speed and for attitude control.

5 Agility: the dive at 35 m/s

5.1 Why the mid speed is the worst speed

A quadrotor changes its direction when it tilts. It tilts when its motors apply different thrusts. The thrust of the weakest motor cannot be less than zero. Thus, the available torque increases with the total thrust [5].

In a dive, gravity and drag both act along the flight path. Thus, the motors supply only a small total thrust. At 35 m/s in a 10° dive, the minimum total thrust is approximately 7 m/s^2. This is less than the thrust necessary to hover.

At the same speed, a change of target acceleration turns the thrust axis fastest. At 55 m/s, the drag is approximately 24 m/s^2. The total thrust is then large, and the attitude changes are small. Table 4 shows the result.

Table 4: Torque per motor in a 10° dive. Reference quadrotor, jerk 5.

	20 m/s	35 m/s	55 m/s
Available	2.8 N	**2.4** N	7.6 N
Necessary	1.5 N	**6.1** N	1.5 N

5.2 Effect of each parameter

Table 5 shows that yaw torque is the limit. Roll torque and pitch torque are not the limit. Our method locks the frame to the side direction of the target. Thus, a change of thrust axis causes a yaw acceleration that increases with the bank angle.

Two changes correct this problem. The first change is a larger yaw reaction coefficient. Use propellers and motors with more reaction torque per newton, or tilt the motors. The second change is a lower yaw inertia. Put the mass near the centre.

A longer arm helps only roll and pitch. More thrust does not help, because the problem is the low minimum thrust.

Table 5: Certified v_{max} in m/s. Dives to 10°, jerk 5.

Parameter	Values	Certified
Yaw coefficient (m)	0.015 / 0.02 / 0.03	30.9 / 33.4 / **52.1**
Inertia factor	0.7 / 1.0 / 1.4	**52.1** / 33.4 / 30.1
Arm length (m)	0.15 / 0.18 / 0.22	33.2 / 33.4 / 33.5
Thrust-to-weight ratio	4 to 8	33.4 for each value

5.3 Less drag gives worse dives

An airframe with less drag dives worse. At $c = 0.006$, the certified speed for dives to 5° at jerk 5 decreases from 52.1 m/s to 39.3 m/s. The cause is the lower total thrust in a dive. Drag is the air brake of the quadrotor. In a pursuit, this air brake keeps the torque available. A smaller rotor-disc area has the same effect (Section 8).

5.4 Wings make the problem worse

We made a prototype of a quadrotor with wings. Its arms had a cambered section that gives lift in the direction of the camera. The lift decreases the load on the motors.

In a dive at 35 m/s, the minimum motor thrust decreased to approximately zero. The available torque decreased to 0.01 N to 0.28 N, but the necessary torque was 0.52 N. For all airframes, the bank angle changes by approximately 5.9° for each m/s^2 of side acceleration. Wings help only with control surfaces, and with these surfaces the vehicle is not a quadrotor.

6 Method: copy the target path from below

Our method has four parts. The planner uses a three-tap filter that is exact for constant acceleration. This filter gets the target positions with a delay of four frames at 240 Hz. The quadrotor flies the filtered target path, at a fixed distance straight below it. The thrust axis follows the planned acceleration, and the yaw follows the side direction of the target. The hinge points the camera straight up, in the limits of its range.

We tested five other methods. Table 6 gives the results. No method was better than our method.

Table 6: Other pursuit methods and their results.

Method	Result
Linear filters with lag correction	With a delay of four frames, the filter must predict. This makes the filter gain $\sum \lvert h_i \rvert$ approximately 2 or more. Thus the planned jerk and snap stay larger than the jerk and snap of the target.
Nonlinear planner with a jerk limit	Against bad targets, the lag was 39 m to 325 m and 13 m/s to 35 m/s. The target turns back when the planner is almost at the target.
Free yaw, mean or larger formations	These methods decrease the torque, but they increase the bank angle, the thrust or the lag.
Weave to decrease rotor-disc drag	For this weave, the motors must supply thrusts from −34 N to +78 N (Section 4).
Wings	The motors stop in dives (Section 5).

The proofs also show a negative result. No proof can force a torque of practical size on the quadrotor. The detection range is 570 m and the lens is 120°. With these values, the quadrotor can respond in 7 s to 30 s. This forces a body rate of only 0.002 to 0.007 rad/s.

Thus, the agility limits come from our tight formation, not from physics. A looser formation can use range and field of view as a margin. But we did not find a looser formation that gives better results.

Put your money and effort into hardware margin and into a clean planner at a high rate. Do not put them into complex path design [6].

7 Rules for operation

Use these rules in operation:

1. Fly below and behind the target, at a distance of 60 m to 100 m. Get into this position before the target gets to its top speed.
2. Expect the most difficult condition at mid speed in a dive. A target that dives at 30 m/s to 40 m/s and starts a turn uses the most margin.
3. Know the two target movements that no quadrotor can follow. The first movement is level flight above the speed limit, which is 57.7 m/s for the reference quadrotor. The second movement, in the isotropic model, is a vertical descent faster than $\sqrt{g/c}$. This speed is 35 m/s at $c = 0.008$.
4. Do not follow from above at a height of more than $D \sin e^*(v)$. If you do, the quadrotor goes past the target.

5. Treat the planner rate as a design value. The proved bounds use 120 Hz to 480 Hz with a delay of four frames. A slower rate increases the change between frames.

8 Design checklist

Table 8 gives the design values for a target at 55 m/s. We tested the recommended set of changes as one unit (Table 7). This set has a thrust-to-weight ratio of 7, $k_n = 0.005$, a yaw coefficient of 0.03 and 30% less inertia.

The parameters have an effect on each other. Without less inertia, the certified speed for dives to 10° at jerk 5 decreases to 35.5 m/s. The cause is the smaller k_n, which also decreases the total thrust in a dive. Thus, always test the full set of changes, not each change alone. Above approximately 59 m/s in 10° dives, the tilt limit of 85° becomes the limit.

Table 7: Recommended set of changes, tested with the proved bounds. Values in m/s.

	Dives to 5°	Dives to 10°
Proved limit	73.9	73.9
Certified, jerk to 3	68.2	58.6
Certified, jerk to 5	68.1	58.4

Table 8: Design checklist for a target at 55 m/s.

	Item	Design value	Reason
1	Thrust-to-weight ratio	7 or more (6 or more with low k_n)	Approximately 5 m/s for each unit
2	Rotor-plane area	k_n not more than 0.005 m^{-1}	Limit increases as $k_n^{-1/4}$
3	Yaw authority	coefficient 0.03 m or more, or tilted motors	Steep dives from 33 to 52 m/s
4	Yaw inertia	not more than 0.014 kg m^2 at 1.5 kg	Same effect as item 3
5	Camera	top, rear; hinge 30° to 70°, or fixed at 50° with a 120° lens	The camera is then not a limit
6	Planner	240 Hz, delay of four frames, exact three-tap filter	Certified configuration
7	Motor reserve	1 to 2 $\mathrm{m/s}^2$ of total thrust at top speed	The top motor gets to its limit first
8	Parasitic drag	c not less than approximately 0.008 m^{-1}	Drag keeps the torque in dives
9	Wings	none	Wings stop the motors in dives
10	Tilt limit	85° or more	Limit above 59 m/s in steep dives

9 What to measure before you use these numbers

Table 9 gives the quantities to measure on your quadrotor. Each quantity has a large effect on the results.

10 Conclusion

Hardware margin, not the pursuit method, limits a quadrotor that follows a fast fixed-wing aircraft. The thrust-to-weight ratio and the rotor-plane area set the top speed. The proved speed limit is less than 1 m/s from steady flight. Yaw authority and yaw inertia set the agility in a dive at mid speed. The camera position removes the camera as a limit. A companion paper gives the formal proofs.

Table 9: Quantities to measure.

Quantity	Method	Reason
k_n and c	Wind tunnel or coast-down test at different pitch angles. Measure the drag along and normal to the rotor plane.	These values set the limit to the power $1/4$.
Thrust and airspeed	Thrust stand in a wind tunnel	The model keeps the maximum motor thrust constant. At $50\,\mathrm{m/s}$, the airflow changes it.
Yaw reaction coefficient	Torque stand	This value sets the dive performance.
Inertia	Bifilar pendulum	Same reason
Target jerk and snap	Flight records of the target type	Each certified speed is for one jerk value.

Reproducibility

The repository is `https://github.com/AbdullahAlmanei/vision-tracking-lean`. Use these commands to get the results again:

- Proofs: `lake build`, then `scripts/axcheck.sh master_sufficient3 p3_bracket_power`.
- Parameter tables: `scripts/paper_sweeps.py`.
- Bounds: `scripts/gap_v3.py` and `scripts/gap_v3.py kn`.
- Weave study: `scripts/explore/weave.py`.

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